

A token is a special permission message required to enter the critical section.

4. What is Critical Section?

Critical section is the part of the program where shared resources are accessed.

5. Which process can enter the critical section?

Only the process holding the token can enter the critical section.

6. How does the token move?

The token moves sequentially from one process to another in a ring structure.

Example:

$P_0 \rightarrow P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow P_4$

7. Why is Token Ring called a distributed algorithm?

Because multiple distributed processes communicate and share the token.

8. What happens if a process does not have the token?

It must wait until the token arrives.

9. What is the purpose of the token?

To avoid multiple processes entering the critical section simultaneously.

10. What is deadlock?

Deadlock is a condition where processes wait forever for resources.

11. Does Token Ring Algorithm avoid deadlock?

Yes

Because only one token exists.

12. What happens if the token is lost?

The system stops working until a new token is generated.

13. Advantages of Token Ring Algorithm

- No deadlock
 - Fair process scheduling
 - Simple implementation
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14. Disadvantages of Token Ring Algorithm

- Token loss problem
 - Delay in large networks
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15. What is process synchronization?

Coordination between processes to avoid conflicts.

16. What is distributed system?

A distributed system is a collection of multiple computers/processes working together.

17. Why mutual exclusion is important?

To prevent data inconsistency and resource conflicts.

18. Which language is used in your program?

Java

19. What is the output of your program?

Token passing from P0 to P1

Token passing from P1 to P2

Process P2 is in Critical Section

20. Real-life applications of Token Ring Algorithm

- Network communication
 - Distributed databases
 - Resource sharing systems
 - Operating systems
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21. What is ring topology?

A network structure where each process/node connects in circular form.

22. What happens after a process exits critical section?

The token is passed to the next process.

23. Can two processes enter critical section together?

No

Because only one token exists.

24. What is starvation?

Starvation means a process waits indefinitely for resources.

25. Does Token Ring Algorithm prevent starvation?

Yes

Because the token circulates fairly among all processes.